Quantitative Data Analysis – Exercises

(Week 12)

In these exercises, you will learn:

- to simulate data
- to perform random sampling and derive measures from samples
- to perform an unpaired two-sample t-test
- to perform a one-way Analysis of Variance (ANOVA)
- to statistically analyze contingency tables using the Chi-squared test
- to perform a correlation analysis including a significance test

In the data analysis process model, these exercises cover part of the steps "Statistical data analysis and/or Modeling" and "Evaluation & Interpretation" (see figure 1). Results of the exercises must be uploaded as separate files (**no .zip files!**) by each student on Moodle. Details on how to submit the results can be found in the tasks below.

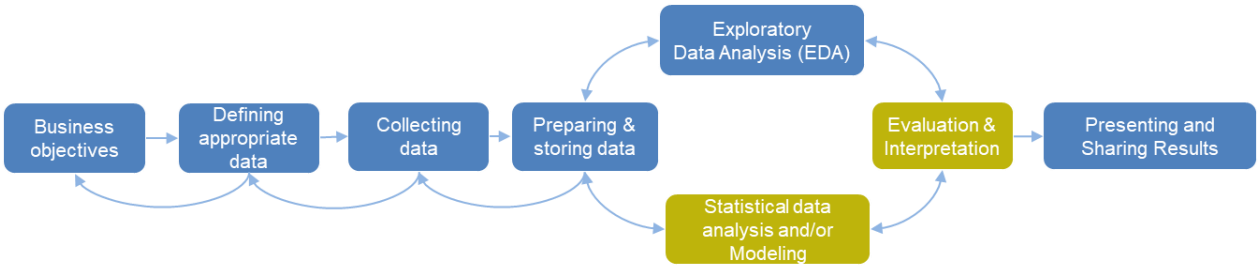


Figure 1: Data analysis process model (see slides of week 07)

Task 1

In this exercise, you will learn what's the difference between a population and a sample. For this, you will use random sampling and bootstrapping. The tasks are:

a) Run the Jupyter notebook 'population_and_samples.ipynb' step by step.
b) Find an own practical example with realistic mean and standard deviation. This could be, for example, diameters of trees in a forest, fuel consumptions of cars in a country, weights of fish in a pond, prices of mobile phones, etc.
c) Run the Jupyter notebook with your example.
d) In the single sample and bootstrap section, vary the sample size and/or number of samples and see how the shape of the histogram changes.

Expected results:

- The Jupyter notebook as html-file 'population_and_samples.html' with the changes made according to b), c) and d)

Task 2

In this exercise, you will learn to perform an unpaired two-sample t-test. The tasks are:

a) Run the Jupyter notebook 'unpaired_two_sample_t-test.ipynb' step by step.
b) Add a section 'Compare rental apartment prices in the cities of Winterthur and Zuerich' to the Jupyter notebook.
c) Create two subsets of the rental apartments data frame. The first subset must contain the prices per m2 of rental apartments in the city of Winterthur. The second subset must contain the prices per m2 of rental apartments in the city of Zürich. Use the following code to create the subsets:

```
df_winti = df.loc[df['bfs_name'] == 'Winterthur']['price_per_m2']
df_zueri = df.loc[df['bfs_name'] == 'Zürich']['price_per_m2']
```

d) Perform an unpaired two-sample t-test based on the two samples to investigate, whether the prices per m2 of rental apartments differ between the cities of Winterthur and Zuerich. Use a significance level of 5%. Use Welch's t-test as in the Jupyter notebook, i.e. stats.ttest_ind(df_winti, df_zueri, equal_var=False).
e) Interpret the results of the statistical test in the notebook.

Expected results:

- The Jupyter notebook as html-file 'unpaired_two_sample_t-test.html' with the changes made according to b), c), d) and e).

Task 3

In this exercise, you will learn to perform a one-way ANOVA. The tasks are:

a) Run the Jupyter notebook 'analysis_of_variance_ANOVA.ipynb' step by step.
b) Create three subsets (data frames) of the apartments data frame analog to task 2, e.g. df_low = df.loc[df['pop_dens'] <= 600]. Use the columns price_per_m2, area and rooms of these subsets in c), d) and e). The subsets must include municipalities with:

   - `low population density (pop_dens <= 600)`
   - `moderate population density ((df['pop_dens'] > 600) & (df['pop_dens'] <= 1500))`
   - `high population density (pop_dens >1500)`

c) Perform a one-way ANOVA based on the three subsets to investigate, whether the prices per m2 of rental apartments differ between the three groups of municipalities. Define a null and an alternative hypothesis. Choose a significance level to decide whether the null hypothesis can be rejected or not. Interpret the results of the statistical test in the notebook. Check the ANOVA assumptions

(Shapiro-Wilk and Levene test) and compare the result with Welch's ANOVA and the Kruskal-Wallis test, as shown in the Jupyter notebook (this also applies to d) and e)).

d) Perform a one-way ANOVA based on the three subsets to investigate, whether the **areas** of rental apartments differ between the three groups of municipalities. Define a null and an alternative hypothesis. Choose a significance level to decide whether the null hypothesis can be rejected or not. Interpret the results of the statistical test in the notebook.

e) Perform a one-way ANOVA based on the three subsets to investigate, whether the **number of rooms** of rental apartments differ between the three groups of municipalities. Define a null and an alternative hypothesis. Choose a significance level to decide whether the null hypothesis can be rejected or not. Interpret the results of the statistical test in the notebook.

Expected results:

- The Jupyter notebook as html-file 'analysis_of_variance_ANOVA.html' with the changes made according to b), c), d) and e).

Task 4

In this exercise, you will learn to statistically analyze a contingency table using the Chi-squared test. The tasks are:

a) Run the Jupyter notebook 'analysis_of_contingency_tables.ipynb' step by step.

b) Look at the Chi-squared test for the contingency table showing the number of apartments for different living area and price categories. The Chi-Square Test is commonly used to test statistical association between two categorical variables.

c) A survey of men and women on the subject of housing preferences produced the results provided in Table 1. Perform a Chi-squared test to investigate whether the differences in preferences are statistically significant (significance level 5%).

Table 1: Results of online survey on housing preferences

	I prefer to live in a single-family house	**I don't prefer to live in a single-family house**
Men	120	86
Women	96	105

Expected results:

- The Jupyter notebook as html-file 'analysis_of_contingency_tables.html' with the changes made according to c)

Task 5

In this exercise, you will learn to perform correlation analyses. The tasks are:

a) Run the Jupyter notebook 'correlation_analysis.ipynb' step by step.

b) Investigate the correlations between population density (variable pop_dens) and price per m^2.
c) Investigate the correlations between percentage of foreigners (frg_pct) and price per m^2.

Expected results:

- The Jupyter notebook as html-file 'correlation_analysis.html' with the changes made according to b) and c) (correlation coefficient, p-value, scatterplot and a short interpretation)